

Rover Route

AI path planning for a Mars rover on real Martian terrain.

The Idea

Choose a start point and a destination on real Mars elevation data. Rover Route plans the safest drive, avoiding steep slopes and rough ground the way rover drivers do, and shows it on a map. Grok Voice plays Mission Control and narrates the drive; Grok Imagine renders a “rover camera” view at each waypoint. You can compare your planned route to the path the real rover actually drove.

Track Fit

Built with Cursor	Pathfinding, data pipeline, and UI all built in Cursor.
Grok Voice / Imagine	Imagine for waypoint views, Voice for Mission Control narration.
Real space data	Mars elevation models (MOLA global, HiRISE high-resolution), real rover routes.
Navigation theme	Autonomous navigation on another planet. Matches the prompt “How do you train AI to better navigate the world?”
Also eligible	Big Red, Software, Design.

Features

MVP

- Load one pre-processed terrain tile (e.g. around Jezero Crater) as a heightmap.
- Compute slope per cell; mark cells above a safety threshold as impassable.
- A* search with a cost based on distance, slope, and roughness.
- Draw the route on a shaded-relief map; click to set start and goal.
- Grok Voice narrates the route summary (distance, max slope, hazards).

Stretch

- 3D terrain view in three.js with the rover driving the path.
- Overlay the real Perseverance traverse for comparison.
- Grok Imagine “rover cam” image at each waypoint.
- Tunable risk slider: fastest route vs. safest route.

Tech Stack and Data

- Data prep: Python with `rasterio`/GDAL to crop and downsample elevation GeoTIFFs.
- Backend: Python (FastAPI) running A*, or A* in the browser on a small grid.
- Frontend: React with a canvas or Leaflet map; three.js for 3D.

- Data: NASA PDS / HiRISE digital terrain models, MOLA global elevation, published rover traverse maps.
- AI: Grok Imagine and Grok Voice APIs.

Limitations and Risks

- **Large, awkward data files.** HiRISE terrain models are large GeoTIFFs in Mars-specific map projections. Processing them can eat hours.
Mitigation: Pre-process one small tile Friday night and ship it as a static file. Do not support arbitrary regions in the MVP.
- **Simplified physics.** Slope and roughness are only part of real rover planning (wheel slip, sand, rocks smaller than the pixel size, power, communication windows).
Mitigation: Say so in the pitch: “a simplified model of how JPL plans drives.”
- **Grok Imagine is not real data.** Generated rover views are artistic guesses, which may clash with the “real space data” spirit of the track.
Mitigation: Label them as artist’s impressions, or ground them by sending real HiRISE imagery of that spot as a reference.
- **Pathfinding performance.** A* on a full-resolution grid (millions of cells) can be slow in the browser.
Mitigation: Downsample the grid, or run A* on the backend.
- **Less “live” demo.** It’s a map-based demo; less physical than pointing a phone at the sky.
Mitigation: Make the 3D drive animation and voice narration the hero moment.
- **Grok API unknowns.** API details, rate limits, and image generation time are not verified.
Mitigation: Pre-generate waypoint images for the demo route and cache them.

Scorecard

Demo-able	4/5	Strong visual if the 3D view works.
Buildable in 24h	3/5	Data prep is the bottleneck.
Navigation theme	5/5	Autonomous navigation on Mars.
Wow factor	4/5	Real Mars terrain impresses.
Technical risk (5 = riskiest)	3/5	Moderate (GIS data, 3D).

Verdict

Strong second choice. Best for a team with some Python and data experience. Reduce risk by fixing on one terrain tile early.